



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - JAN/FEB 2023

SUBJECT : CSE 2154 - OBJECT ORIENTED PROGRAMMING (MAKEUP)

OBJECT ORIENTED PROGRAMMING [CSE 2154]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

- 1A) What are varargs? Illustrate varargs and the ambiguity associated with them with an example program. (3)
- 1B) Write two different Java programs to illustrate the following:
i) Using labelled break as a Form of goto. (3)
ii) Labelled break inside nested loops.
- 1C) Create a **Date** class with three integer instance variables named **day**, **month**, **year**. It has a constructor with three parameters for initializing the instance variables, and it has one method named **daysSinceJan1()**. Override to **toString()** method for **Date** class to be used to display date in the format dd/mm/yyyy. It computes and returns the number of days since January 1 of the same year, including January 1 and the day in the **Date** object. For example, if **day** is a **Date** object with **day** = 1, **month** = 3, and **year** = 2000, then the call **date.daysSinceJan1()** should return 61 since there are 61 days between the dates of January 1, 2000, and March 1, 2000, including January 1 and March 1. Include a **DateDemo** class that tests the **Date** class by creating an array of **Date** class of size n. Don't forget leap years. (4)
- 2A) Create a generic class with one type parameter, T. Provide an appropriate constructor that initialises the data member of this class that holds a reference to an array of type T. Also, implement the following methods:
1) **getDevice(int id)** that returns a **Device** object whose id is passed in method argument. If no record is found, it displays an appropriate message and returns the first device.
2) **sort()** that sorts the records of **Device** based on **watt** where, class **Device** contains two data members **watt** and **id**. Create two classes **Fan** and **Kettle** that are derived from **Device**. **sweep** (short) and **brand** (String) are data members of **Fan**. Also, **Kettle** contains two data members **capacity** (double) and **name** (String). Provide a method **display()** in both the classes to print the details on the console output. Write a driver class that
i) creates an array of **Fan** and **Kettle** devices.
ii) Instantiate the generic class on **Fan** and **Kettle** devices and display them.
iii) Invoke the **getDevie(int id)** method on both the instances and display the device details.
iv) Use the instances to sort the records by calling the **sort()** method. (5)
- 2B) Demonstrate the use of bounded wildcard arguments. (3)
- 2C) Demonstrate with an example how the CERT rule to validate return values of a method is beneficial. (2)

3A) Create a Book class with private instance variables for Book ID, Title, Author, and Price. Add appropriate accessor methods for these variables. Then create a subclass Novel with instances variables for the Category and Publisher. Also, include appropriate constructors for your classes. Create a class with main() method to instantiate an array of FIVE Novels. Initialize the array during creation and display all the Novel's Titles published by 'Pearson' Publishers. (5)

3B) Create a package testpkg consisting of two classes A and B, class A being public should have three variables, three methods (each variable and method of different access specifier: default, public and protected) and a member interface consisting of two methods. Class B being default should have a single method. Name this program as A.java. (3)
Write a Java program to compile and execute the package testpkg using imports.

3C) Consider the following code. Is it written correctly? If so, mention output of the program, else, list the error/errors in the program. (2)

```

interface X
{
    int x=10;
    void methodX(int k);
}
class Y implements X
{
    void methodX(int k) {
        x=k;
        System.out.println("Method X:"+x);
    }
}

```

4A) "Synchronized threads communication, is termed as inter-thread communication" Justify your Answer with proper code snippets how do Threads communicate. (2)

4B) Write a Java Program to Sort an Array of Integers (Read from user randomly) in the Ascending order and Descending Order simultaneously. Analyse your program with proper Input. (3)

4C) Create a javafx application select Insurance Policy and pay the premium as shown in the given Figure. Write the complete program with import statements and Analyse the program with proper inputs? (5)

5A) Demonstrate with a suitable example how to use a Toggle Button and Check box in JavaFx and Analyse your code with proper inputs? (4)

- 5B) What is Dynamic method dispatch? Demonstrate with a suitable example how java implements run-time polymorphism. (3)
- 5C) What are Packages? Analyse hierarchy of packages with suitable example? (3)

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